Table 6.3 Spin I, parity (+ even, − odd), nuclear magnetic moment (μ n.m.) and quadrupole moment $\hat{Q}$ (10^{-28} m^2) for some nuclides ^{A}X

Stable nuclides				Radioactive nuclides				
^{A}X	I (±)	μ_I	$\hat{Q}$	^{A}X	I (±)	μ_I	$\hat{Q}$	$t_{½}$
^{1}H	1/2 +	+2.793		^{3}H	1/2 +	+2.979		12.33 y
^{2}H	1 +	+0.857	+0.003	^{14}C	0 +			5730 y
^{10}B	3 +	+1.801	+0.085	^{24}Na	4 +	+1.690		14.959 h
^{11}B	3/2	+2.689	+0.041	^{32}P	1 +	−0.252		14.262 d
^{12}C	0 +			^{36}Cl	2 +	+1.285	−0.018	3.01×10^{5} y
^{13}C	1/2	+0.702		^{45}Ca	7/2	−1.327	+0.046	163.8 d
^{14}N	1 +	+0.404	+0.019	^{55}Fe	3/2			2.73 y
^{16}O	0 +			^{60}Co	5 +	+3.799	+0.44	5.271 y
^{17}O	5/2 +	−1.894	−0.026	^{64}Cu	1 +	−0.217		12.70 h
^{19}F	1/2 +	+2.629		^{95}Zr	5/2 +			64.02 d
^{23}Na	3/2 +	+2.218	+0.101	^{131}I	7/2 +	+2.742	−0.40	8.0207 d
^{31}P	1/2 +	+1.132		^{137}Cs	7/2 +	+2.841	+0.051	30.0 y
^{33}S	3/2 +	+0.644	−0.076	^{140}La	3	+0.730	+0.094	1.678 d
^{39}K	3/2 +	+0.391	+0.049	^{198}Au	2	+0.593	+0.68	2.6952 d
^{59}Co	7/2	+4.627	+0.404	^{232}Th	0 +			1.405×10^{10} y (α)
^{87}Sr	9/2 +	−1.094	+0.335	^{235}U	7/2	−0.38	+4.55	7.038×10^{8} y (α)
^{141}Pr	5/2 +	+4.275	−0.059	^{238}U	0 +		13.9	4.468×10^{9} y (α)
^{197}Au	3/2 +	+0.146	+0.547	^{239}Pu	1/2 +	+0.203		2.411×10^{4} y (α)

This form makes it more obvious why γ is referred to as the gyromagnetic ratio. Equation (6.29) is valid both for angular momentum and spin, but of course the value is different for electrons and protons. The angular velocity ω_γ of the precession is found to be

$$\omega_\gamma = \mu \overrightarrow{B} / \overrightarrow{p}_{\text{rot}} \tag{6.30}$$

By replacing angular velocity with frequency ($\omega = 2\pi\upsilon$)

$$\upsilon_\gamma = \gamma \cdot \overrightarrow{B} / 2\pi \overrightarrow{p}_{\text{rot}} \tag{6.31}$$

or

$$\upsilon_\gamma = \gamma \cdot \overrightarrow{B} / 2\pi \tag{6.32}$$

where υ_γ is the Larmor precession frequency.

6.3. THE SINGLE-PARTICLE SHELL MODEL

6.3.1. Quantum number rules

Let us assume that a nucleon moves around freely in the nuclear potential well, which is spherically symmetric, and that the energy of the nucleon varies between potential and

kinetic energy like a harmonic oscillator, i.e. the potential walls (see Figs. 6.1 and 6.2) are parabolic. For these conditions the solution of the Schrödinger equation yields:

$$E(\text{nucleon}) = \hbar\left(2\,U_o/mr^2\right)^{1/2}[2(n-1)+l] \tag{6.33}$$

where U_o is the potential at radius $r = 0$ and m is the nucleon mass. We have defined n and l previously. The square root, which has the dimension s^{-1}, is sometimes referred to as the oscillator frequency (ω in Table 6.6). The following rules are valid for nucleons in the nuclear potential well:

(a) l can have all positive integer values beginning with 0, independent of n;
(b) the energy of the l state increases with increasing n as given by (6.33);
(c) the nucleons enter the level with the lowest total energy according to (6.33) independent of whether n or l is the larger;
(d) there are independent sets of levels for protons and for neutrons;
(e) the Pauli principle is valid, i.e. the system cannot contain two particles with all quantum numbers being the same;
(f) the spin quantum numbers must be taken into account (not included in (6.33)).

Following these rules, the nucleons vary greatly in energy and orbital motion. However these rules do not exclude the existence of two nucleons with the same energy (so-called degenerate states), provided the quantum numbers differ

6.3.2. Nuclei without nucleon spin-orbit coupling

If we calculate the sequence of energy levels on the assumption that n is constant, we obtain the pattern in Table 6.4. This sequence correspond exactly to the electronic states in atoms but it does not agree with the observed magic numbers 2, 8, 20, 28, 50, etc., for nuclei.

Applying the rules of the previous section a new set of nucleon numbers is obtained: 2, 8, 20, 40, 70, etc. (Table 6.5, and left column Table 6.6). This level scheme allows a large amount of degeneracy. For example, for $n = 1$ and $l = 3$ (1f-state) we find from (6.33) that $[2(n-1)+l] = 3$, which value is obtained also for $n = 2$ and $l = 1$ (2p-state).

Table 6.4 Energy levels derived on basis of permitted values for the azimuthal quantum number *l* and spin quantum number s

l	State	Possible quantum values	Number of states (including spin)	Accumulated nucleon number
0	s	0	$1 \times 2 = 2$	2
1	p	−1,0,+1	$3 \times 2 = 6$	8
2	d	−2,−1,0,+1,+2	$5 \times 2 = 10$	18
3	f	−3,−2,−1,0,+1,+2,+3	$7 \times 2 = 14$	32
4	g	−4,−3,−2,−1,0,+1,+2,+3,+4	$9 \times 2 = 18$	50
l	-	$-l$, …, $+l$	$2(2l+1)$	–

Table 6.5 Energy levels according to equation (6.25)

Levels	Number of nucleons	Accumulated nucleons
1s	2	2
1p	6	8
1d and 2s	10 + 2	20
1f and 2p	14 + 6	40
1g, 2d, and 3s	18 + 10 + 2	70
1h, 2f, and 3p	22 + 14 + 6	112

Since the f-state can have 14 and the p-state 6 nucleons, the degenerate level of both states can contain 20 nucleons. However, the numbers still do not correspond to the experimental magic numbers.

A further refinement is possible if we assume that the nuclear potential well has straight walls, i.e. the potential energy $U(r)$ is $-U_o$ at $r < r_n$ while it is infinite at $r \geq r_n$, where r_n is the nuclear radius. This assumption, when introduced (see Fig. 6.2 and figure in Table 6.6) in the Schrödinger-equation, leads to a splitting of the degenerate levels, so that the lowest energy is obtained for the state with lowest main quantum number n. For our example of the 1f and 2p-states the 1f orbitals are lower in energy than the 2p. This refinement yields the middle row of levels in Table 6.6 but still does not lead to the correct magic numbers.

6.3.3. Nuclear level scheme with nucleon spin-orbit coupling

In multielectron atoms, the Russell-Saunders coupling is present in light atoms. However, in the heaviest atoms of many electrons and in highly charged nuclei, the j—j (spin-orbit) coupling better describes the systems. Haxel, Jensen, Suess, and Goeppert-Mayer in 1949 suggested that the nucleons always experience a strong spin-orbit coupling according to (6.12)

$$j = l \pm s \tag{6.34}$$

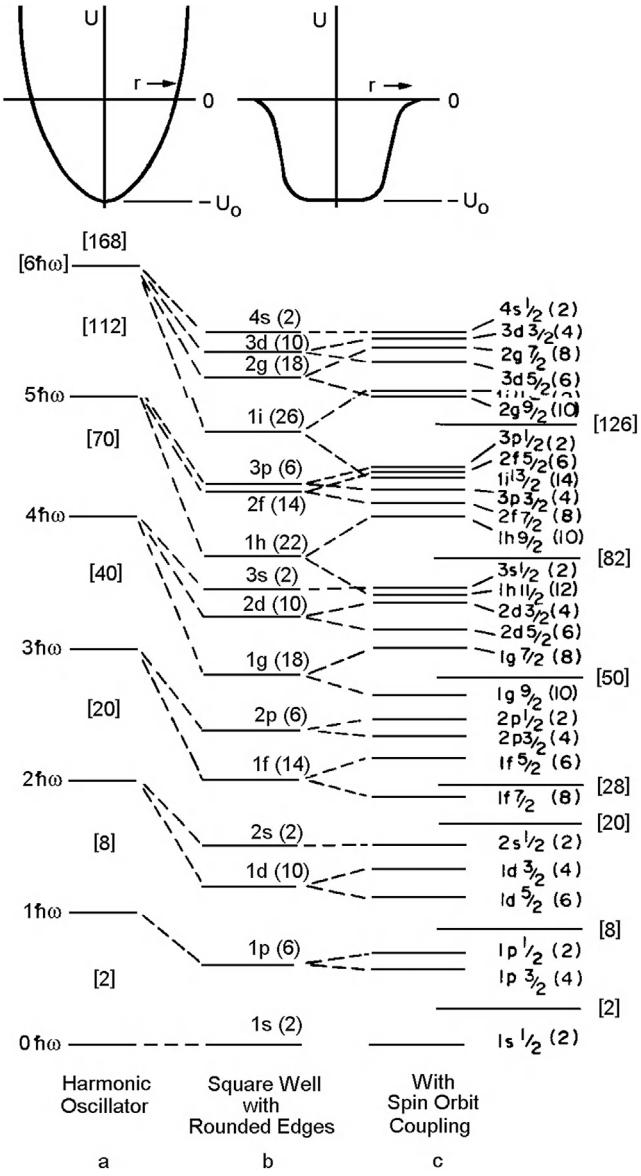
and that the total spin of the nucleus I is the sum of the nucleon spins

$$I = \Sigma j \tag{6.35}$$

In this way, all I levels are split into two levels with quantum values $l + ½$ and $l - ½$, of which the former has the lowest energy value (the opposite of the electron case). This yields the row of levels on the right in Table 6.6. Because of the energy splitting the new levels group together so they fit exactly with the experimental magic numbers.

As an example, consider the level designation $1i_{11/2}$. This has the following interpretation: the principal quantum number is 1; i indicates that the orbital quantum

Table 6.6 Nuclear energy levels obtained for the single-particle model by solving the schrödinger-equation (a) for a harmonic oscillator and rounded square-well potentials (b) without and (c) with spin-orbit coupling. The $n\hbar\omega$ figures to the left are energies for model a, according to eqn (11.25). numbers in ( ) indicate orbital capacities and those in [ ] give cumulative capacity up to the given line



From Gordon and Coryell.

number l is 6; the angular momentum quantum number j is 11/2 ($j = l - 1/2$). The number of permitted nucleons in each level is $2j + 1$, thus 12 for $j = 11/2$.

The magic numbers correspond to sets of energy levels of similar energy just as in the atom the K, L, M, etc., shells represent orbitals of similar energy. The N-electronic shell contains the 4s, 3d, 4p sets of orbitals while the 4th nuclear "shell" contains the $1f_{5/2}$, $2p_{3/2}$, $2p_{1/2}$, and $1g_{9/2}$ sets of orbitals. Remember that there are separate sets of orbitals for protons and for neutrons.

6.3.4. The nuclear spin

The nuclear spin I is obtained from (6.34) and (6.35). Since j is always a half-integer, nuclides with odd number of nucleons (odd A) must have odd spin values (odd I), while those with even A must have even I. The nucleons always pair, so that even numbers of protons produce no net spin. The same is true for even numbers of neutrons. For nuclei of even numbers for both N and Z, the total nuclear spin I is always equal to zero. Some ground state nuclear spins are given in Table 6.3.

For odd A the nuclear spin is wholly determined by the single unpaired nucleon (single particle model). Let us take the nucleus $^{13}_{6}C$ as an example. It contains 6 protons and 7 neutrons. From Table 6.6 we conclude that there are 2 protons in the $1s_{1/2}$ level and 4 protons in the $1p_{3/2}$ level. A similar result is obtained for the first 6 neutrons but the 7th neutron must enter the $1p_{1/2}$ level. The value of I for $^{13}_{6}C$ is thus ½. Another example is $^{51}_{23}V$, which has 23 protons and 28 neutrons. Since $N = 28$, the neutrons do not contribute to I. From Table 6.6 we see that we can accommodate 20 protons in the orbitals $1s^2$, $1p^6$, $1d^{10}$, $2s^2$. The next 3 protons must go into the $1f_{7/2}$ level ($1\,f^3_{7/2/}$, where, however, 2 of them are paired. Therefore, the single unpaired proton has $j = 7/2$, leading us to predict $I = 7/2$, which also is the measured nuclear spin value.

When a nucleus is excited, either through interaction with other particles or in a decay process, for nuclei with N or Z values near magic numbers the paired nucleons seem not to be perturbed by the excitation (if it is not too large). As a result, we can associate the excitation with any unpaired nucleons. Let us choose the decay of ^{47}Ca to ^{47}Sc (see Fig. 6.5). ^{47}Ca has 27 neutrons and in the ground state the last 7 neutrons must occupy the $1f_{7/2}$ level. The ground state of (the unstable) ^{47}Sc has 21 protons, the unpaired proton can only be accommodated in the $1f_{7/2}$ level. Thus both of the ground states have $I = 7/2$. The next higher energy states for ^{47}Sc involve the levels $1f_{5/2}$ and $2p_{3/2}$. In the Figure it is seen that these levels are observed although their order is reversed from that in Table 6.6. This reflects some limitation of the single-particle model, and is explained in §6.5.

For odd-odd nuclei the nuclear spin is given by

$$I(\text{odd-odd}) = j_p + j_n = (l_p \pm ½) + (l_n \pm ½) \tag{6.36}$$

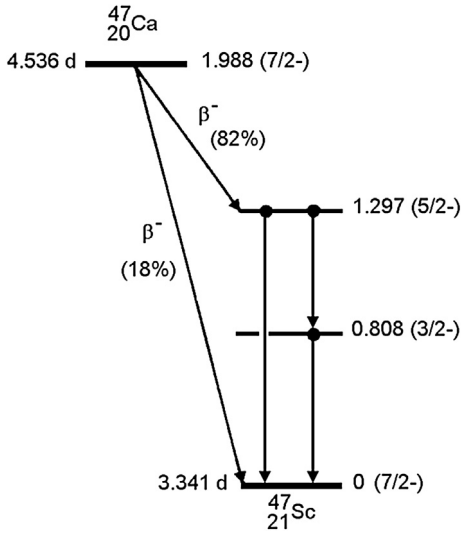


Figure 6.5 Part of the decay scheme for $A = 47$; the data are experimental values.

According to the rules formulated by Brennan and Bernstein (useful in the range $20 < A < 120$ for ground states and low-lying long-lived isomeric states), if the odd particles are both particles (or both holes) in their respective unfilled subshells then[1]

$$\begin{aligned} &\text{if } \ j_\mathrm{p} + j_\mathrm{n} + l_\mathrm{p} + l_\mathrm{n} \ \text{ is even}, \quad &&\text{then } I = \left|j_\mathrm{p} - j_\mathrm{n}\right| \\ &\text{if } \ j_\mathrm{p} + j_\mathrm{n} + l_\mathrm{p} + l_\mathrm{n} \ \text{ is odd}, \quad &&\text{then } I = \left|j_\mathrm{p} \pm j_\mathrm{n}\right| \end{aligned} \tag{6.37}$$

but if we have both particles and holes, $\qquad$ then $I = j_\mathrm{p} + j_\mathrm{n} - 1$

The use of these rules can be illustrated by the case of Cu which has its odd proton in the $1f_{5/2}$ orbital and its odd neutron in the $2p_{3/2}$ orbital. Thus for the proton, $j = 5/2$, $l = 3$; for the neutron, $j = 3/2$, $l = 1$. Because $j_\mathrm{p} + j_\mathrm{n} + l_\mathrm{p} + l_\mathrm{n}$ is even, we use $I = |5/2 - 3/2| = 1$, which is the observed spin value. The lightest nuclei are exceptions to this rule since they often exhibit LS coupling according to (6.18). ^{10}B is an example; it has 5 protons and 5 neutrons, the fifth nucleon being in the $1p_{3/2}$ state. Thus $I = j_1 + j_2 = 3$, which is observed.

We mentioned in §6.2.4 that theoretical calculations of the nuclear magnetic moment, (6.21), are usually not satisfactory. The value of μ(nucleus) can have a number of values depending on m_I, with a maximum value of I. The parity[2] of the nucleus follows the rules: (i) when both particles are in states either of even parity or of odd parity,

1 Vertical bars used here denote absolute value.

2 Parity is the behaviour of wave functions when all coordinate signs are reversed. Even parity — no effect, else odd parity.

they combine to a system of even parity. (ii) one particle in a state of even parity and one in a state of odd parity combine to a system of odd parity.

6.4. DEFORMED NUCLEI

6.4.1. Deformation index

Both the liquid-drop model and the single-particle model assume that the mass and charge of the nucleus are spherically symmetric. This is true only for nuclei close to the magic numbers; other nuclei have distorted shapes. The most common assumption about the distortion of the nuclide shape is that it is ellipsoidal, i.e. a latitudinal cross-section of the nucleus is an ellipse. Figure 6.6 shows the oblate (flying-saucer-like) and prolate (egg-shaped) ellipsoidally distorted nuclei; the prolate shape is the more common. Deviation from the spherical shape is given by

$$\beta = 2(a - c)/(a + c) \tag{6.38}$$

where a and c are the elliptical axes as shown on in Figure 6.6(c). For prolate shape $\beta > 0$, and for oblate shape $\beta < 0$. The maximum deformation observed is about $\beta = \pm 0.6$.

The deformation is related to the nuclear shell structure. Nuclei with magic numbers are spherical and have sharp boundary surfaces (they are "hard"). As the values of N and Z depart from the magic numbers the nucleus increases its deformation.

6.4.2. Electric multipoles

In a spherical nucleus we assume the charge distribution to be spherical and the nucleus acts as a monopole. In the deformed nuclei, the nuclear charge has a non-spherical distribution. The potential at a point, $\bar{x}$, $\bar{y}$, $\bar{z}$ (Fig. 6.6) will be found to vary depending on the charge distribution and mode of rotation of the nucleus. The nuclear charge may be distributed to form a dipole, a quadrupole, etc. Nuclei are therefore divided into

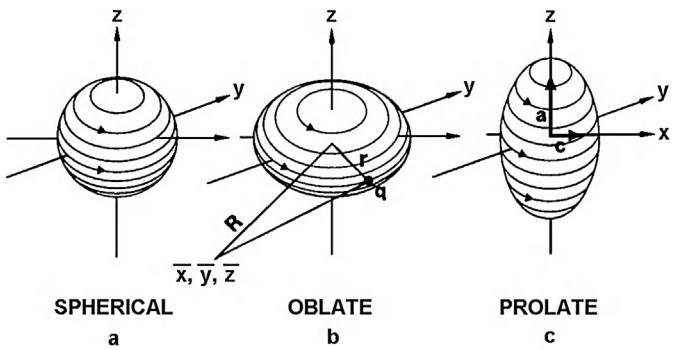


Figure 6.6 Three nuclidic shapes: (a) spherical nucleus, (b) oblate (extended at the equator), and (c) prolate (extended at the poles).